AN EFFECTIVE METHOD BASED ON HUMAN DAILY ACTIVITIES TO DETERMINE THE RATE OF DEPRESSION

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Abstract-Stress is one of the biggest problems in today's fast-paced society, and because people's lifestyles change so quickly, it frequently results in depression. It can be challenging to diagnose depression, especially when its symptoms coexist with those of other medical condition Investigating personality-based characteristics that contribute to student depression was the aim of this study. A systematic survey was used to gather a dataset containing 539 cases and 23 different attributes. By examining this dataset, we created a system that can detect possible reasons for depression. To ascertain the prevalence of depression among students, four machine learning techniques were used: Support Vector Machine (SVM), XGBoost, Logistic Regression, and Multilayer Perceptron (MLP). Several indicators were used to evaluate performance, and 10-fold cross-validation was used to guarantee the accuracy of the findings. Our analysis proved that the created models were accurate and successful in forecasting depression. XGBoost (92.25%), Multilayer Perceptron (91.10%), Logistic Regression (90.50%), and Support Vector Machine (92.80%) all had the highest accuracy. Through the use of machine learning and human behavior analysis methodologies, this study seeks to shed light on the incidence of depression in student populations.

Keywords: Support Vector Machine Algorithm, XG Boost Algorithm, Multilayer Perceptron (MLP), Data Mining, Depression, and Logistic Regression Algorithm.

I. INTRODUCTION

Depression is a condition that can appear in a variety of ways. Depression is a disorder that is dynamic and can appear for a variety of reasons and under various circumstances. Brain injuries and extreme mental stress can be caused by depression. According to WHO estimates, depression impacts about 300 million people globally. [2] To determine the root cause of depression in people, we apply data mining approaches. Over the past 20 years, data mining techniques have been used more and more in a variety of fields [3]. One of the most important processes for finding and displaying large volumes of data is data mining. It offers a path to information discovery as well. Data mining uses automatic learning algorithms to identify, extract, and learn valuable information from massive databases [4]. Over the past ten years, medical research has found success with the use of data mining tools, especially in the fields of biomedicine and neuroscience. Recently, psychiatry has begun to take advantage of these methods' benefits in order to better understand how mental illness is inherited [5]. Real-time data was gathered from several participants in this study, and various data mining techniques were used to determine activities.

II. RELATED WORKS

This study presents a fuzzy logic method for flood detection to promptly notify customers of the possibility of flooding. Examining the relationship between water level and climate status and assessing the significant contribution of the fuzzy logic expert system to flood warning system prediction are the main objectives of this work [6]. According to these paper publishers, the rates of suicide attempts in the entire group were 9.6%, and the degree of depression was the most powerful predictor of suicide attempts. Compared to the non-depressive group, the suicide attempt rates were 2.8 and 5.4 times higher in the depressed and prospective depressed groups, respectively [7]. The analysis of emotions related to depression is done using Natural Language Processing (NLP). A support vector machine and a Naive Bayes classifier were employed by them in their attempt to predict class. They used Twitter to get information. Additionally, the findings were presented using the key categorization criteria, such as confusion and F1-score, matrix, and precision [8]. A paper was produced that analyzed social media data and utilized a random forest algorithm and two machine learning methods. One algorithm will identify subjects who are depressed, and another algorithm will aid in identifying individuals who are not depressed. Their conclusion was that the dual model is more beneficial than the one[9]. Additionally, a researcher used Twitter data to make predictions in a paper published by a publisher, and they also compared the outcome with other publications [10].

III. METHODOLOGY

Although depression analysis has not seen many comparative investigations, it is currently being categorized. Our efforts are guided by these works. This section provides a brief overview of techniques and how they use effective tools and algorithms. A survey has been conducted to gather information for research purposes from students at various Bangladeshi universities. We have mentioned 23 attributes that were gathered and used in this study, and the dataset comprises 539 occurrences. Several approaches have been used in this investigation. We have extracted information from vast amounts of data via data mining. The strategies are as follows:

- Information Gathering
- Preprocessing of Data
- Algorithmic Implementation
- Simulation Environment

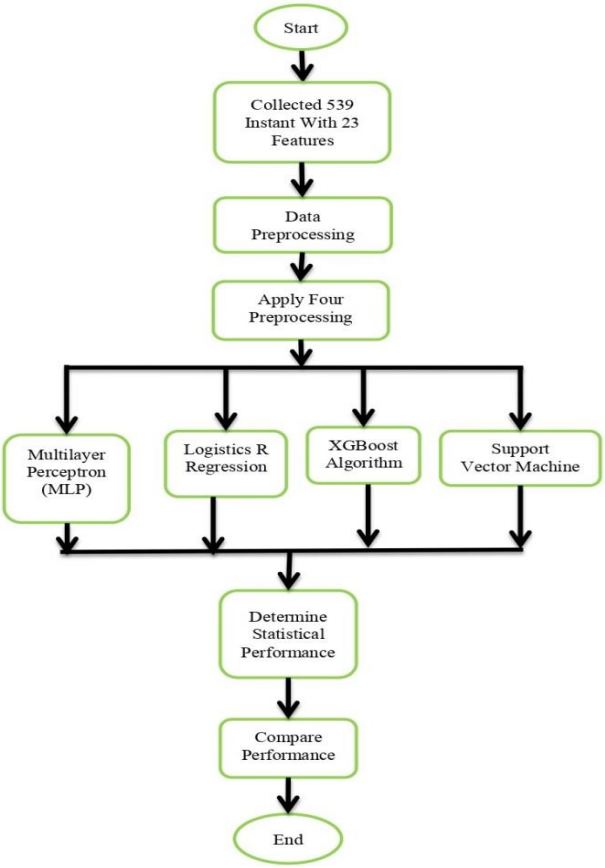

Fig.1: Procedure for System Operation

The flow chart shown in Fig. 1 shows the entire working process.

A. *Information Gathering*

Our questionnaire was created after consulting with many psychiatrists. We gather data from B.Sc., M.Sc., and Undergraduate. Upon receiving the response, we generate our dataset. In the dataset, there are 539 cases. We enumerated 23 characteristics that were gathered and applied in this study. In Table I, the features list is displayed.

B. *Preprocessing of data*

We have addressed and cleaned the missing data using a variety of methods, including addressing data discrepancies, smoothing noisy data, and replacing missing values or rows.

Figure 2 depicts the process of data preparation.

C. *Algorithmic Implementation*

For classification, we employed four different methods. Multilayer Perceptron's (MLP), Logistic Regression Algorithm, XG boost Algorithm, Support vector Machine Algorithm are these.

1) **Multilayer Perceptron** (MLP): An example of a multilayer perceptron (MLP) is the Feed-forward Artificial Neural Network (ANN) class [11]. The term MLP can be used to refer to networks made up of several perceptron layers, or it can be used broadly to refer to any feed-forward ANN. An input layer, an output layer, and a hidden layer are the three node layers that make up an MLP [11].

TABLEI: Features List

Feature	Category	Data Distribution
Age	-	Min: 17, Max: 33
Gender	Male	66.50%
	Female	33.50%
Do you accept all social norms?	Yes	66.03%
	No	33.97%
What are your thoughts on yourself?	Extrovert	57.15%
	Introvert	42.85%
Do you own a smartphone?	Yes	93.10%
	No	6.90%
Do you enjoy hanging out with friends?	Yes	89.25%
Are you content in your current role?	No	10.75%
	Yes	66.02%
	No	33.98%
Are you anxious about your work?	Yes	71.23%
	No	28.77%
Do you talk to someone about your problems?	Yes	61.05%
Do you find solitude comfortable?	No	38.95%
	Yes	61.22%
	No	38.78%
Did you attempt suicide?	Yes	14.83%
	No	85.17%
Do you think suicide is the answer to all your problems?	Yes	12.40%
	No	87.60%
Are you content with your family?	Yes	87.00%
	No	13.00%
Do you feel like a burden to your family?	Yes	38.75%
	No	61.25%
Where are you more at ease?	Family	62.30%
	Friends	29.85%
	Others	7.85%
Family Size	4 Members	36.72%
	5 Members	20.60%
	6 Members	21.76%
	Other	20.92%
Do you engage in extracurricular activities?	Yes	67.52%
	No	32.48%
Do you find your education system challenging?	Yes	67.30%
	No	32.70%
Daily Sleep Duration	6 Hours	18.92%
	7 Hours	27.80%
	8 Hours	25.90%
	Other	27.38%
SSC Result	5.00	46.55%
	4.50	7.40%
	4.00	5.00%
	Other	41.05%
HSC Result	5.00	21.51%
	4.50	8.52%
	4.00	12.42%
	Other	57.55%
University CGPA	4.00	2.21%
	3.25	2.23%
	3.00	12.83%
	Other	82.73%
Do you suffer from depression? (Class Label)	Yes	40.62%
	No	59.38%

2) ***XGBoost Algorithm:*** An approach for supervised learning called XGBoost predicts a target variable by combining the estimates of simpler models. To accelerate model discovery, XGBoost makes use of distributed and parallel computing, together with a novel tree learning technique. XGBoost uses the weighted quartile sketch algorithm to handle sparse data sets[14].

3) ***Logistic Regression Algorithm:*** Based on a

collection of input factors, the machine learning method known as logistic regression predicts the likelihood of a binary result using a mathematical technique. In logistic regression, a linear combination of input features is converted into a probability value between 0 and 1 using the logistic function. There are various forms of logistic regression, such as ordinal, multinomial, and binary, depending on the type of dependent variable [1].

4) ***Support Vector Machine Algorithm:*** A supervised machine learning approach called a support vector machine (SVM) classifies and regresses data using statistical learning theory.

SVMs maximize the margin between classes in order to identify a hyper plane that divides them in a high-dimensional space. SVMs are frequently used to solve classification issues including image recognition and text categorization. They are also useful for outlier detection and regression[16].

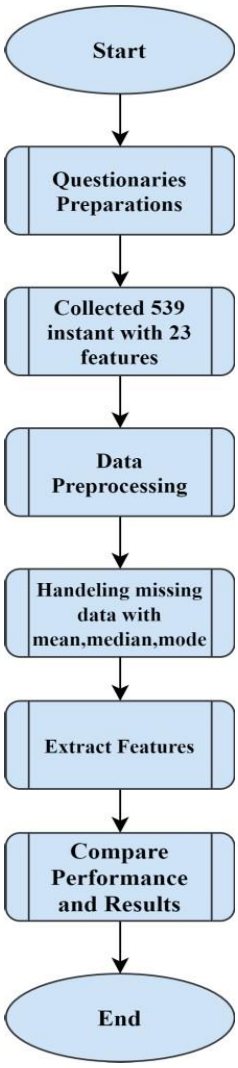

Fig2: Data Preparation Procedure

D. Simulation Environment

- Waikato Environment for Knowledge Analysis (WEKA) Version: 3.8.5

IV. EXPERIMENTED OUTCOMES AND ANALYSIS

A. Accuracy

The accuracy can be defined as:

$$\text{Accuracy} = \frac{TRPV + TRNV}{TRPV + TRNV + FLPV + FLNV}$$

Table II-Experimented Outcome

Metric	Multilayer Perceptron (MLP)	Logistic Regression	XG Boost	Support Vector Machine (SVM)
Accuracy (%)	91.10	90.50	92.25	92.80
False Positive Rate	0.988	0.888	0.935	0.985
Average Precision (%)	89.50	88.50	90.80	90.90
Average Recall (%)	90.90	87.77	92.85	90.20
Average F1-Measure (%)	90.40	94.80	89.65	92.90
Average ROC Area (%)	69.40	90.61	50.85	91.45
Average PRC Area (%)	89.55	50.81	88.25	62.80
Confusion Matrix (TP, FP, TN, FN)	**[483 20] [30 7]**	**[500 4] [39 0]**	**[499 5] [36 3]**	**[455 5] [35 2]**

Here, TRPV = True Positive, TRNV = True Negative, FLPV=False Positive, FLNV=False Negative, Positive=PV and Negative=NV.

B. Recall, True Positive Rate (TRPVR), or Sensitivity

A test's true positive rate, or TP, is the likelihood that individuals with the illness will yield true-positive results [14].

$$TRPV\ R = \frac{TRPV\ R}{PV}$$

C. Precision

D. F1-score

Additionally, it is recognized by F-Measure, which is defined as [15].

$$F_1 = 2\ \frac{\text{Precision x Recall}}{\text{Precision + Recall}}$$

E. Discussions

Conversations The results of the four methods are displayed in a bar graph in Figure 3. Table II displays the results of the experiment. Are you depressed for class? We employed the Multilayer Perceptron (MLP), Logistic Regression, XGBoost, and Support Vector Machine Algorithm (Are you depressed?). Accuracy, TP Rate, FP Rate, Precision, Recall, F-Measure, MCC, Roc Area, and PRC Area were obtained by employing three methodologies. The results obtained by applying the Multilayer Perceptron (MLP) algorithm were 94.1 TP Rate, 74.1% FP Rate, 96% Precision, 95% F-Measure, 22.2% MCC, 69.40% ROC Area,

and 89.55 PRC Area. Through the use of the Logistic Regression technique, we were able to obtain 92.8% TP Rate, 99.9% FP Rate, 99.2 Precision, 95.9% F-Measure, 49.5% MCC, 90.61% ROC Area, and 50.81 PRC Area. Using the XGBoost method, we obtained a TP rate of 93.2%, FP rate of 62.5%, precision of 99%, F-measure of 96%, MCC of 49.8%, ROC area of 50.85%, and PRC area of 88.25%. These results were obtained using the Support Vector Machine algorithm: 92.9% TP Rate, 71.4% FP Rate, 98.9% Precision, 95.8% F-Measure, 42.9% MCC, 91.45% ROC Area, and 62.80% PRC Area.

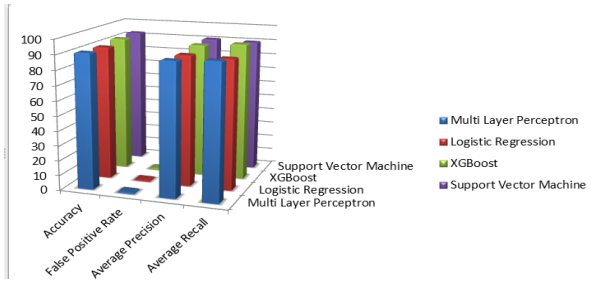

Fig. 3: Bar-chart between MLP, Logistic Regression, XG Boost and Support Vector Machine Algorithm.

V. Conclusion

By examining information obtained from structured surveys, this study sought to determine the causes of depression in college students. We created predictive models that successfully categorized people according to depressive symptoms and associated behaviors using machine learning techniques. Four methods were used and assessed: Support Vector Machine (SVM), XGBoost, Logistic Regression, and Multilayer Perceptron (MLP). With 92.80% classification accuracy, the SVM algorithm outperformed the others, closely followed by XGBoost (92.25%), MLP (91.10%), and Logistic Regression (90.50%). These findings show that machine learning methods have the potential to be used in the early identification and assessment of depression, which might be crucial for prompt intervention and mental health assistance. Nevertheless, there were restrictions when gathering the data. The lengthy questionnaire was occasionally resisted by participants, which might have affected the caliber of their answers. Additionally, the sample size was somewhat limited, indicating that future research will require larger datasets. In our upcoming efforts, we want to:

Increase the size of the dataset by adding more people with a range of backgrounds.

Improve prediction performance by implementing deep learning techniques.

Sort depression severity into three categories for a more detailed analysis: severe, medium, and low.

Examine other algorithms, such Random Forest and Naive Bayes, as well as sophisticated pre-processing methods to raise the accuracy and effectiveness of model.

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